

run-api-app-challenge-matrix.sh

§6.AE / §6.X / §6.AG gate entry point for the :api-app module's Compose UI Challenge tests (Challenge01..Challenge04 under lava.api.app.challenges).

Script: scripts/run-api-app-challenge-matrix.sh

What it does

The :api-app sibling of scripts/run-challenge-matrix.sh. THIN GLUE: it delegates emulator boot + lifecycle to the Containers submodule's cmd/emulator-matrix CLI (per §6.X / §6.AG — the device **MUST** come from a Containers-orchestrated cold-booted emulator AVD, **NEVER** a live/physical ADB device). It forwards --runner=auto (Linux→containerized via /dev/kvm, macOS→host-direct+HVF), the :api-app debug APK, the Challenge test class, the AVD matrix, and the evidence dir; the CLI's exit code is the gate's source of truth.

Usage

```
./scripts/run-api-app-challenge-matrix.sh [--test-class
lava.api.app.challenges.ChallengeNN] \
    [--avds "name:api:form,..."] [--evidence-dir <dir>] [--no-
build] \
    [--boot-timeout <duration>] [--test-timeout <duration>] \
    [--container-image <ref>] [--container-runtime <podman|
docker>]
```

--test-timeout pass-through — LVA-161 (2026-08-26)

emulator-matrix defaults --test-timeout to **10 minutes** (cmd/emulator-matrix/main.go:123), and that budget covers the **TEST step only** — cold boot is timed separately as boot_seconds. This script previously never set it, so every run inherited that default.

The default is far below a real whole-module sweep: a Challenge run is a **single** gradle invocation covering every selected class, and gradle writes its JUnit XML at the **end** of that invocation. A kill at the timeout therefore destroys the results of **every** class, including those that had already passed — the matrix then reports the entire selection as failed when nothing about the product was wrong. The sibling scripts/run-challenge-matrix.sh recorded the measured case: a run killed at test_seconds=600.02 with signal: killed after completing 81 of 104 tests.

This script now accepts --test-timeout <duration> and forwards it verbatim to the CLI when set; when unset, the flag is omitted entirely and the CLI default applies. This script performs **no** sizing of its own — the caller owns that. The pipeline's Challenge phase (scripts/pipeline/phase-02-test-challenge.sh) derives the value as 300s + 45s × <selected class count> and passes it in; an operator invoking this script directly for more than a handful of classes should pass a sized value rather than accept 10m.

A run killed at the timeout is **not** a test failure: test_seconds at the budget plus signal: killed plus an absent JUnit XML is the signature. The honest remedy is to raise --test-timeout or narrow the class selection and re-run for a real verdict.

--boot-timeout is the separate, pre-existing knob for cold-boot headroom on a loaded host (contention, not a product defect); the two budgets do not interact.

Constitutional bindings

§6.AE (per-app Challenge matrix), §6.X (container-submodule emulator wiring), §6.AG (Containers-driven, no live device), §6.AH (VM/container, never host-direct), §6.I (per-AVD attestation).